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An Extensive Study on Statistical Data Anonymization Algorithms

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***Abstract* - Gigantic volume of detailed individual information is constantly gathered and divulging of these information is gainful for data mining application. Datais collected from thedata holders by various data publishers before beingthe release of data to the data beneficiary for the purpose of research analysis and mining. This released data may reveal the private and personal information of individuals. Thus arises the most important research issue of privacy in data publishing. Here, in this paper, we provided the analysis of the existing techniques for statistical data anonymization which are used for privacy preservation in published data along with the efficiency and effectiveness of each. This study will help the researchers to understand variety of different anonymization methods for microdata publishing, relationship between k-values, and anonymization degree and execution time.**

Keywords: Generalization, Privacy preservation publishing, Data Anonymization, statistical anonymization

I. INTRODUCTION

In this digital age of cloud and IOT applications, various organizations gather gigantic volumes of data from heterogeneous data stores that may include records for individuals. For example, many private companies and government agencies collect & use data generally called microdata [1] and also data like personal browsing histories stored by various web search engines [2] [3]. Once data collection is over, data collector often publish this data in manner which is much useful for achieving objectives of data mining & research. However, this published data that mostly contains sensitive personal information and the effusion of the sensitive information infringes the privacy of individuals. Most popular attacks, for example are, finding theMassachusetts governor disease [4], finding an AOL user browsing history [5] etc. Privacy preservation became major target area of research due to such attacks. Privacy preservation is about protection from leaking the identity of any individual.

A. Microdata Publishing

The term 'microdata' commonly refers to the data table which contains information about a particular individual or organization. Microdata could be voter data, medical data, citizen data and demographic data which contains many distant attributes. These attributes are further classified as:

1. *Key Attributes*- These attributes, also called explicit identifiers are liable for identifying there cord owners or individual explicitly. For example, name is key attribute in the table I.

2. *Quasi-Identifier*- These are attributes which can indirectly help in identifying an individual when the values of these attributes are combined. For examples, zipcode, Age, gender in table I.

3. *Sensitive Attribute* –These attributes should not disclose values to attacker. E.g. person's disease or salary of an employee are sensitive attributes.

B. Disclosure Risks

When microdata is published, the necessity for more security to sensitive attribute of the individuals arises. In [7] [8] [9],three types of information disclosures are explained as: attribute, identity, and membership disclosure.

1. *Attribute Disclosure:* This type of disclosure occur due to leakage in new sensitive information.
2. *Identity Disclosure:*If a specific record of published data links to individual, the "anonymity" is broken.
3. *Membership Disclosure:* In this type, the invader checks published data to trace a particular person's record in the published data.

C. Anonymizing Data

The published data might have fruitful information for the data miner, however it can also have individual's identity risk. Thus, the goal of privacy publishing is maximizing the gain with minimal risk of individual's identification. Anonymization can be defined asapproach for preservation of privacy in published data that tries to safeguard the sensitive data and identification of record owners. This may be adepted thru anonymization before data publishing. Firstly,for anonymization, the key attributes are removed from the database. In Table I, since the key attribute 'Name' may tell the patient's identity, it is removed from Table I and the released data is shown in Table II.

TABLE I. MICRODATA

Name	Gender	Zipcode	Age	Disease
Rama	F	713201	41	Heart problem
Meena	F	744608	24	Flu
Ketan	M	744106	27	Heart problem
Aarpan	M	713001	35	Flu
Komal	F	713302	39	Viral infection
Aaina	F	744406	20	Cancer

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TABLE II. PUBLISHED DATAAFTER SENSITIVE ATTRIBUTE REMOVAL

Gender	Zipcode	Age	Disease
F	713201	41	Heart problem
F	744608	24	Flu
M	744106	27	Heart problem
M	713001	35	Flu
F	713302	39	Viral infection
F	744406	20	Cancer

However, an invader might have information about individuals of the published data. Thus simply explicit identifier removal is not enough.The invader can get this knowledge either from own knowledge, or from any public data tables e.g. voter data as shown in Table III. From data Table II and Table III, one can infer conveniently that 'Rama'is suffering from Heart disease.

TABLE III. VOTER MICRODATA

Name	Gender	Zipcode	Age	Voter ID
Rama	F	713201	41	6547382
Meena	F	744608	24	5392922
Ketan	M	744106	27	6235498
Aarpan	M	713001	35	4393855
Komal	F	713302	39	2345129
Aaina	F	744406	20	9824533
Gyan	F	774376	30	6235364
Heena	M	774506	38	4523930
Isha	F	774706	34	7645403

Thus, additional anonymization is needed for prevention from such forms of attacks.

II. RELATED WORK

We can anonymize a table in number of ways before the publishing of it. The main aim of these ways is to suit the specific privacy requirements as well as assuring that the anonymized information remains valuable for analysis.

A. Anonymization Procedures

The most accepted methods of anonymization are grouping-and-breaking and perturbation.

1) *Grouping-and-Breaking:* In this type of operation, the records are divided into many horizontal partitions and then in each partition, the precise relation of the quasi and sensitive attributes is broken[5]. The grouping is done with objective of making the individuals in distinguishable in the same group so that the unique identification of any individual in the group becomes difficult for the invader. The breaking is aimed to dilute the link of quasi and sensitive identifiers, thus the invader gets confused in identification of individual which could be inferred by establishing link between individual record pin-pointed by QI values & sensitive attribute values e.g. HIV. The grouping and breaking techniqueis aditionally categorised into categories: suppression, generalization and bucketization.

- *Suppression:* In this technique [1], either the tuple or an attribute value is removed and replaces the attribute value or complete tuple with symbol "*" which indicates any value can be there. Table IV is obtained after the gender and age attribute suppressed.

TABLE IV. Published data after suppression

Gender	Zipcode	Age	Disease
*	713201	*	Heart problem
*	744608	*	Flu
*	744106	*	Heart problem
*	713001	*	Flu
*	713302	*	Viral infection
*	744406	*	Cancer

Drawbacks of suppression: Reduction in the quality of the data hugely.

- *Generalization:* Samarati and Sweeney[1] [4] suggested the usage of generalization. Generalization reinstate the attribute values in the data set with semantically consistent but less precised value. Thus, several records may have same QI values due to generalized replacement. Generalization reinstates the definite values with a more ordinary description to conceal the details of the attributes which makes the QIDs less identifying. When we have numeric value, it may be reinstated to a range of values. For example, 26 age attribute value may be converted to range 21-30. One categorical value can be reinstated to another categorical value that may denote a broader concept of the original categorical value. Table V is table generated by generalization.

TABLE V. PUBLISHED DATAAFTER GENERALIZATION ON AGE ATTRIBUTE

Gender	Zipcode	Age	Disease
F	713201	41-50	Heart problem
F	744608	21-30	Flu
M	744106	21-30	Heart problem
M	713001	31-40	Flu
F	713302	31-40	Viral infection
F	744406	11-20	Cancer

Drawbacks of generalization: a) *It* misses significant amount of information in case of high-dimensional data,. b) Generalization stirred from bane of dimensionality. c) Reduction in the data utility of generalized data. d) Loss in correlations between various attributes.

- *Bucketization:* The process of bucketization [12] is quite similar to generalization process, but here any QI attribute or sensitive attribute is not modified. Instead, it

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assigns a distinctive ID known as GID to each partition once the records are divided into various partitions. All tuples in one partition will have same GID value. Thus, the two new tables are created, namely quasi attribute (QI) table and the sensitive table. The important point to note here is that the grouping thru bucketization is similar to the grouping thru generalization process, however the bucketized data has all original tuple values whereas generalized data has generalized tuples values. Therefore, bucketization process allows users to use the original values for their research and data analysis. Table VI shows this procedure.

Drawbacks of Bucketization: a) The membership disclosure is not restrained. b) Requires clear demarcation between Quasi-Identifiers and Sensitive Attributes. c) The correlation between the Quasi-Identifiers and Sensitive Attributes break down.

TABLE VI. PUBLISHED DATAAFTER BUCKETIZATION

Gender	Zipcode	Age	GID
F	713201	41	1
F	744608	24	1
M	744106	27	2
M	713001	35	2
F	713302	39	3
F	744406	20	3

(a)QI table

GID	Disease
1	Heart problem
1	Flu
2	Heart problem
2	Flu
3	Viral infection
3	Cancer

(b) Sensitive table

2) Perturbation

In this operation [13], an arbitrary value replaced any value in the data table. For example, in gender attribute, 'M' is converted to 'F' and vice versa. The perturbation example is shown in Table VII.

TABLE VII. PUBLISHED DATA AFTER PERTURBATION

Gender	Zipcode	Age	Disease
M	713201	41	Heart problem
M	744608	24	Flu
F	744106	27	Heart problem
F	713001	35	Flu
M	713302	39	Viral infection
M	744406	20	Cancer

Drawbacks of perturbation: Data utility is reduced.

B. Privacy models

As discussed in section I,the three types of attributes are key attributes, quasi attributes and sensitive or confidential attributes. The term privacy has two views –anonymity i.e. non-identification of any individual from the published data and confidentiality i.e. released data access should not disclose secret information about specific individual. The privacy models are there to fulfil privacy requirements and to implementit the data transformation is performed to uphold information utility. The majority of privacy models targets the blocking of three cynical dangers to privacy of individuals namely attribute disclosure, identity disclosure and membership disclosure. The individual's identity may be revealed if the invader knows the quasi identifier of targeted individual. Besides, transformation of data is enforced thru anonymization procedures as discussed in earlier section. Many privacy models are already suggested to safeguard individual privacy concerns due to the consideration of the various attacking cases of the data, The relation of different privacy models with different attacks is summarized in table VIII below:

TABLE VIII. RELATION BETWEEN PRIVACY MODELS AND ATTACKS

Privacy Model/ Attacks	Attibute Disclosure	Identity Disclosure	Memebership disclosure
K-anonymization		Yes	
MultiR k-Anonymity		Yes	
l-Diversity		Yes	
t-Closeness	Yes		
ϵ-Differential Privacy			Yes

C. SLICING

To solve the problems occurring in generalization and bucketization, slicing [15]is introduced as new technique for the privacy preservation of published data. In slicing technique, we doslicing on partitioned data where in partition can be made horizontally and/or vertically. The hugely correlated attributes are clustered into one column in vertical partitioning, where in, the column contains group of hugely correlated attributes. However, in horizontal partitioning, the tuples are clustered into number of buckets. The values of column are arbitrary sorted to break the linking between different columns. For example, table IX shows sliced data part of table II.

TABLE IX. SLICED DATA FOR PUBLISHING

Zipcode, Age	Gender, Disease
713201, 41 744608, 24 744106, 27	F, Heart problem F, Flu M, , Heart problem
713001, 35 713302, 39 744406, 20	M, Flu F, Viral infection F, Cancer

Slicing reduces the dimensionality of the data. When compared to generalization and bucketization, slicing retains

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better data utility. Slicing groups together hugely correlated attributes maintaining correlation among attributes and thus breaking down the coalition of attributes which are uncorrelated, thereby providing better privacy of published data. The identification of these attributes is easy chore as they are not rare. Slicing offers better protection of privacy since tuples have multiple matching tuples.

Advantage of slicing: a) Slicing, as compared with generalization, gives better data utility. b) Slicing is extra convincing than bucketization. c) Slicing accords with high dimensional data also.

III. k- ANONYMITY ALGORITHMS

Various algorithms exists based on diverse k-anonymity models for achieving k-anonymity. In this paper, we will discuss three well-known algorithms: Samarati's Algorithm, Sweeney's Algorithmand Incognito Algorithm which uses the generalization and suppression [4], [10], [18] procedures.

A. Samarati's Algorithm

The algorithm given by Samarati, finds thepotential k-anonymous answers by hopping between various levels in domain generalization hierarchy (DGH). It provides solution in less time using the binary search. The AG_TS model is implemented by this algorithm and thus to achieve k-anonymity, suppression can be used. This algorithm makes assumption that the solutions which results in table data with minimal generalizations are the best solutions [15].

B. Sweeney's Algorithm (Datafly Algorithm)

This algorithm looks for the solutions which are attained when variables are generalized by most unique values as the best solutions. This technique finds the solution by traversing very few nodes in the lattice. Therefore, it is very efficient approach from the time aspect, and hence the name Datafly[9, 17]. In simple words, for every node of the lattice, checking the attribute which has most unique items and then that attribute is moved up one level as per the corresponding hierarchy. Repeating it till there are less than k- rows which are not adhering to k-anonymity and then remaining rows are supressed.

C. Incognito Algorithm

This algorithm gives maximum attainable k-anonymous full-domain generalizations of relation keeping threshold in tuple suppression. It starts with inspecting quasi-identifiers having single- attribute subsets, and then repeating regularly while inspecting k-anonymity for quasi-identifier larger subsets [18].

Table X shows the comparative analysis of these three algorithms on various parameters

TABLE X. Comparison of Algorithms

Parameter/Algorithm	**Samarati**	**Sweeney**	**Incognito**
Node evaluation	All nodes are evaluated at some generalization level	Skips a lot of nodes when moving between levels	All set of feasible solutions are generated
Anonymization method used	Solution is provided having minimum Generalization and Suppression.	The solution doesn't contain minimum generalization.	Solution is generated with minimum generalization and Suppression.
Execution Time	More execution time.	Execute very quickly on large data sets.	Execution time is more.
Optimal Solution	The optimal solution is not necessarily located.	Optimal solution is not necessarily located.	Always locates the optimal solution
Information Loss	Min Information loss	Max Information loss	Information loss is minimal
Advantage	• The output may have the possibility of becomingan optimal solution • Results are much better than Sweeney. • Uses binary search to obtain solution	Very few nodes are checkedfor k-anonymity, thus it is giving results very fast.	• The algorithm always locates the optimal solution as per specified criteria. • It finds all the k-anonymous full domain generalizations.
Disadvantage	• With the variation in k, the possibility to get optimal solution dramatically varies. • Maximum Suppression lattice size.	• The resulting data is too much generalized as algorithm jump over many nodes. Due to this, sometimes released data having little information, is not suitable for researches. • It suppresses all values within tuple.	The algorithm takes lot of time for traversing the solution space as it uses bottom-up breadth first search method.

IV. CONCLUSION

Data anonymization is important feature of Microdata disclosures. Also, in published data, privacy is important challenge as the imprudent revelation of some data assets will damage the prospects. There exist many approaches for data anonymization like generalization, suppression, bucketization and perturbation which are used for preservation of privacy in published data. However, generalization loses significant amount of data in case of high dimensional data and damaged badly due to bane of dimensionality. Membership disclosure is not preserved in bucketization since it requires clear demarcation of quasi attributes and Sensitive attributes. Quality of data is reduced terribly in Suppression. Slicing is recent technique for anonymization in preservation of privacy in publish data where in the data can be partitioned horizontally and vertically. Slicing decreases the dimensionality of data but provides preservation of utility. Also, the correlations between highly correlated attributes are maintained. As compared to generalization, better data utility is provided by slicing. Slicing has greater effectiveness as

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compared to bucketization and can handle high-dimensional data, providing attribute disclosures and membership disclosures. Also, the anonymization level is precisely proportionate to the number of records and to fill the gap between the released data and its privacy level, we must choose k carefully wherein thealgorithm selection is significant in maintaining the Anonymization-Privacy Preserving ratio.

REFERENCES

[1] P. Samaratiand L. Sweeney, "Protecting privacy when disclosing information:-anonymity and its enforcement through generalization and suppression," SRI International, SRI-CSL-98-04, 1998.

[2] Shweta Sharma Hitesh Gupta PriyankJain "A Study Survey of Privacy Preserving Data Mining". International Journal of Research & Innovationin Computer engineering, Vol 2, Issue 2, April 2012.

[3] Pingshui WANG. "Survey on Privacy Preserving Data Mining". International Journal of Digital Content Technology and its Applications Volume 4, Number 9, December 2010.

[4] L. Sweeney, "K-Anonymity: A model for protecting privacy," International Journal on Uncertainty, Fuzziness and Knowledge-based Systems, vol. 10, no. 5, pp. 557–570, 2002.

[5] M. Barbaroand T. Zeller, "Afaceis exposed for AOL searcher no. 4417749," New York Times, 2006.

[6] A. Narayanan and V. Shmatikov, "Robust de-anonymization of large sparse datasets," in Proceedings of the IEEE Symposium on Security and Privacy (S&P), pp. 111–125, 2008.

[7] G. T. Duncan and D. Lambert, "Disclosure-limited datadissemination," Journal of The American Statistical Association, pp. 10–28, 1986.

[8] D. Lambert, "Measures of disclosure risk and harm," Journal of Official Statistics, vol. 9, pp. 313–331, 1993.

[9] M. e. Nergiz, M. Atzori, and C. Clifton, "Hiding the presence of individuals from shared databases," in Proceedings of the ACM SIGMOD International Conference on Management of Data (SIGMOD), pp. 665– 676, 2007.

[10] C. Aggarwal, "On k-Anonymity and the Curse of Dimensionality," Proc. International Conf. Very Large Databases (VLDB), pp. 901-909, 2005.

[11] D. Kiferand J. Gehrke, "Injecting Utility into Anonymized Data Sets," Proc. ACM SIGMOD International Conf. Management of Data (SIGMOD), pp. 217-228, 2006.

[12] X. Xiao and Y. Tao, "Anatomy: Simple and effective Privacy Preservation," Proceeding International Conf. Very Large Data Bases (VLDB), pp. 139-150, 2006.

[13] Sinha, B. K., and J. Kumar. Privacy Preserving Clustering in Data Mining. Diss. 2010.

[14] Seema Kedar, Sneha Dhawale, Wankhade Vaibhav, Pavan Kadam , Siddharth Wani , Pavan Ingale, "Privacy Preserving Data Mining", International Journal of Advanced Research in Computer and Communication engineering Vol. 2, Issue 4, April 2013

[15] Mynavathi, R., N. Sowmiya, and D. Vanitha. "Survey of Various Techniques to Provide Multilevel Trust in Privacy Preserving Data Mining."

[16] Jian-min, Han, Cen Ting-ting, and Yu Hui-qun. "An improved V-MDAV algorithm for l-diversity." Information Processing (ISIP), 2008 International Symposiums on. IEEE, 2008.

[17] J. Soria-Comas, J. Domingo-Ferrer, D. Sanchez, and S. Martınez, "Improving the Utility of Differentially Private Data Releases via k-Anonymity", In Proceedings of the 12th IEEE International Conference on Trust, Security and Privacy in Computing and Communications, TRUSTCOM-13, pages 372–379, 2013.

[18] L. Sweeney, "Datafly: a system for providing anonymity in medical data. In Database Security", XI: Status and Prospects, IFIP TC11 WG11.3 11th International Conference on Database Security, 356-381, 1998.

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